

AGGREGATE FUNCTIONS with WHERE, GROUP BY, HAVING, ORDER BY

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STEP 1: CREATE TABLE

The screenshot displays the SQL Developer environment. The left sidebar shows a tree of schemas including bankdb, bookflow_db, bookflow_week2_db, college_yt, practice, sql_clg, and sys. The main editor window shows a SQL script with the following content:

```
1 • CREATE DATABASE IF NOT EXISTS aggregate_db;
2 • USE aggregate_db;
3 • CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     branch_name VARCHAR(50),
7     transaction_type VARCHAR(20),
8     amount DECIMAL(10,2),
9     transaction_date DATE
10 );
```

The bottom panel shows the 'Output' window with the 'Action Output' tab selected. It contains a table with the following data:

#	Time	Action	Message
✓ 42	21:14:02	SELECT * FROM physician ORDER BY em...	9 row(s) returned
✓ 43	21:14:22	SELECT * FROM physician ORDER BY em...	9 row(s) returned
✓ 44	21:14:48	SELECT MAX(employeeid) FROM physicia...	1 row(s) returned
⚠ 45	21:23:47	CREATE DATABASE IF NOT EXISTS aggr...	1 row(s) affected, 1 warning(s): 1007 Can't
✓ 46	21:24:03	USE aggregate_db	0 row(s) affected
✓ 47	21:24:13	CREATE TABLE bank_transactions (txn...	0 row(s) affected

STEP 2: INSERT DATA

Navigator

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_dlg
- sys

Administration Schemas

Information

No object selected

Object Info Session

SQL File 6

```

3 • CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     branch_name VARCHAR(50),
7     transaction_type VARCHAR(20),
8     amount DECIMAL(10,2),
9     transaction_date DATE
10 );
11 • INSERT INTO bank_transactions VALUES
12 (101, 'Sai Akhil', 'Hyderabad', 'Deposit', 5000, '2024-01-05')
13 (102, 'Sita', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
14 (103, 'Kiran', 'Vijayawada', 'Deposit', 11000, '2024-01-08'),
15 (104, 'Anil', 'Vizag', 'Deposit', 8000, '2024-01-10'),
16 (105, 'Priya', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
17 (106, 'Ramesh', 'Hyderabad', 'Deposit', 15000, '2024-01-12'),
18 (107, 'Keerthi', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
19 (108, 'Rahul', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
20 (109, 'Sneha', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
21 (110, 'Madhu', 'Vijayawada', 'Deposit', 11000, '2024-01-16');

```

Limit to 1000 rows

Output

Action Output

#	Time	Action	Message
✓ 43	21:14:22	SELECT * FROM physician ORDER BY em...	9 row(s) returned
✓ 44	21:14:48	SELECT MAX(employeeid) FROM physicia...	1 row(s) returned
⚠ 45	21:23:47	CREATE DATABASE IF NOT EXISTS aggr...	1 row(s) affected, 1 warning(s): 101
✓ 46	21:24:03	USE aggregate_db	0 row(s) affected
✓ 47	21:24:13	CREATE TABLE bank_transactions (txn...	0 row(s) affected
✓ 48	21:25:15	INSERT INTO bank_transactions VALUES...	10 row(s) affected Records: 10 D

SUM()

SCHEMAS

Filter objects

bankdb

bookflow_db

bookflow_week2_db

college_yt

practice

sql_clg

sys

Administration

SCHEMAS

Information

No object selected

Object Info

Session

Limit to 1000 rows

18 (107, 'Keerthi', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'

19 (108, 'Rahul', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),

20 (109, 'Sneha', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),

21 (110, 'Madhu', 'Vijayawada', 'Deposit', 11000, '2024-01-16');

22

23 • SELECT SUM(amount) AS Total_Amount

24 FROM bank_transactions;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Total_Amount
▶	69500.00

Result 1 x

Output

Action Output

#	Time	Action	Message
✓ 44	21:14:48	SELECT MAX(employeeid) FROM physicia...	1 row(s) returned
⚠ 45	21:23:47	CREATE DATABASE IF NOT EXISTS aggr...	1 row(s) affected, 1 warning(s): 1007 (
✓ 46	21:24:03	USE aggregate_db	0 row(s) affected
✓ 47	21:24:13	CREATE TABLE bank_transactions (txn...	0 row(s) affected
✓ 48	21:25:15	INSERT INTO bank_transactions VALUES...	10 row(s) affected Records: 10 Dupli
✓ 49	21:25:43	SELECT SUM(amount) AS Total_Amount F...	1 row(s) returned

AVG()

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

Administration Schemas

Information

No object selected

Limit to 1000 rows

```
19 (108, 'Rahul', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
20 (109, 'Sneha', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
21 (110, 'Madhu', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
22
23 • SELECT SUM(amount) AS Total_Amount
24 • FROM bank_transactions;SELECT AVG(amount) AS Average_Transaction_Amount
25 FROM bank_transactions;
```

Result Grid

Average_Transaction_Amount
6950.000000

Result 2 x

Output

Action Output

#	Time	Action	Message
45	21:23:47	CREATE DATABASE IF NOT EXISTS aggregate_db	1 row(s) affected, 1 warning(s): 1007
46	21:24:03	USE aggregate_db	0 row(s) affected
47	21:24:13	CREATE TABLE bank_transactions (txn_id INT(4) NOT NULL AUTO_INCREMENT, customer VARCHAR(50) NOT NULL, location VARCHAR(50) NOT NULL, transaction_type VARCHAR(20) NOT NULL, amount INT(10) NOT NULL, transaction_date DATE NOT NULL, PRIMARY KEY (txn_id))	0 row(s) affected
48	21:25:15	INSERT INTO bank_transactions VALUES (108, 'Rahul', 'Hyderabad', 'Deposit', 9000, '2024-01-14')	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
49	21:25:43	SELECT SUM(amount) AS Total_Amount FROM bank_transactions	1 row(s) returned
50	21:26:03	SELECT AVG(amount) AS Average_Transaction_Amount FROM bank_transactions	1 row(s) returned

MAX()

Navigator

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

WEEK1*

ddl

WEEK2

SQL File 5*

SQL File 6* x

SQL File 7*

JOINS

Limit to 1000 rows

21

(110, 'Madhu', 'Vijayawada', 'Deposit', 11000, '2024-01-16');

22

23

24

25

26

27

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Highest_Transaction
15000.00

Administration

Schemas

Information

No object selected

Result 3 x

Output

#	Time	Action	Message	Duration / Fetch
✓ 46	21:24:03	USE aggregate_db	0 row(s) affected	0.000 sec
✓ 47	21:24:13	CREATE TABLE bank_transactions (txn...	0 row(s) affected	0.031 sec
✓ 48	21:25:15	INSERT INTO bank_transactions VALUES...	10 row(s) affected Records: 10 Duplicates:...	0.016 sec
✓ 49	21:25:43	SELECT SUM(amount) AS Total_Amount F...	1 row(s) returned	0.000 sec / 0.
✓ 50	21:26:03	SELECT AVG(amount) AS Average_Transa...	1 row(s) returned	0.000 sec / 0.
✓ 51	21:26:22	SELECT MAX(amount) AS Highest_Transa...	1 row(s) returned	0.000 sec / 0.

MIN()

Navigator

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

WEEK1* ddl WEEK2 SQL File 5* SQL File 6* x SQL File 7* JOINS

23 • SELECT SUM(amount) AS Total_Amount

24 • FROM bank_transactions;SELECT AVG(amount) AS Average_Transaction

25 FROM bank_transactions;

26 • SELECT MAX(amount) AS Highest_Transaction

27 FROM bank_transactions;

28 • SELECT MIN(amount) AS Lowest_Transaction

29 FROM bank_transactions;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Lowest_Transaction
1000.00

Administration Schemas

Information

No object selected

Object Info Session

Result 4 x

Output

Action Output

#	Time	Action	Message	
✓ 47	21:24:13	CREATE TABLE bank_transactions (txn...	0 row(s) affected	0
✓ 48	21:25:15	INSERT INTO bank_transactions VALUES...	10 row(s) affected Records: 10 Duplicates:...	0
✓ 49	21:25:43	SELECT SUM(amount) AS Total_Amount F...	1 row(s) returned	0
✓ 50	21:26:03	SELECT AVG(amount) AS Average_Transa...	1 row(s) returned	0
✓ 51	21:26:22	SELECT MAX(amount) AS Highest_Transa...	1 row(s) returned	0
✓ 52	21:26:47	SELECT MIN(amount) AS Lowest_Transac...	1 row(s) returned	0

COUNT()

Navigator

CHEMAS

- Filter objects
- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

WEEK1* ddl WEEK2 SQL File 5* SQL File 6* SQL File 7*

25 FROM bank_transactions;

26 • SELECT MAX(amount) AS Highest_Transaction

27 FROM bank_transactions;

28 • SELECT MIN(amount) AS Lowest_Transaction

29 FROM bank_transactions;

30 • SELECT COUNT(*) AS Total_Transactions

31 FROM bank_transactions;

Result Grid

Total_Transactions
10

Filter Rows: Export: Wrap Cell Content:

Result 5 x

Output

Action Output

#	Time	Action	Message
✓ 48	21:25:15	INSERT INTO bank_transactions VALUES...	10 row(s) affected Records: 10 Du
✓ 49	21:25:43	SELECT SUM(amount) AS Total_Amount F...	1 row(s) returned
✓ 50	21:26:03	SELECT AVG(amount) AS Average_Transa...	1 row(s) returned
✓ 51	21:26:22	SELECT MAX(amount) AS Highest_Transa...	1 row(s) returned
✓ 52	21:26:47	SELECT MIN(amount) AS Lowest_Transac...	1 row(s) returned
✓ 53	21:27:07	SELECT COUNT(*) AS Total_Transactions ...	1 row(s) returned

Administration Schemas

Information

No object selected

Object Info Session

WHERE + SUM()

SCHEMAS

Filter objects

bankdb

bookflow_db

bookflow_week2_db

college_yt

practice

sql_clg

sys

Administration

Schemas

Information

No object selected

Object Info

Session

Limit to 1000 rows

28

•

SELECT MIN(amount) AS Lowest_Transaction

29

FROM bank_transactions;

30

•

SELECT COUNT(*) AS Total_Transactions

31

FROM bank_transactions;

32

•

SELECT SUM(amount) AS Total_Deposit

33

FROM bank_transactions

34

WHERE transaction_type = 'Deposit';

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Total_Deposit
59000.00

Result 6

×

Output

Action Output

#	Time	Action	Message
49	21:25:43	SELECT SUM(amount) AS Total_Amount F...	1 row(s) returned
50	21:26:03	SELECT AVG(amount) AS Average_Transa...	1 row(s) returned
51	21:26:22	SELECT MAX(amount) AS Highest_Transa...	1 row(s) returned
52	21:26:47	SELECT MIN(amount) AS Lowest_Transac...	1 row(s) returned
53	21:27:07	SELECT COUNT(*) AS Total_Transactions ...	1 row(s) returned
54	21:27:24	SELECT SUM(amount) AS Total_Deposit F...	1 row(s) returned

GROUP BY

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_dlg
- sys

Administration Schemas

Information

No object selected

Object Info Session

```

32 • SELECT SUM(amount) AS Total_Deposit
33 FROM bank_transactions
34 WHERE transaction_type = 'Deposit';
35 • SELECT branch_name,
36        SUM(amount) AS Total_Amount
37 FROM bank_transactions
38 GROUP BY branch_name;

```

Result Grid

branch_name	Total_Amount
Hyderabad	34500.00
Vijayawada	23000.00
Vizag	12000.00

Result 7

Output

Action Output

#	Time	Action	Message
✓ 50	21:26:03	SELECT AVG(amount) AS Average_Transa...	1 row(s) returned
✓ 51	21:26:22	SELECT MAX(amount) AS Highest_Transa...	1 row(s) returned
✓ 52	21:26:47	SELECT MIN(amount) AS Lowest_Transac...	1 row(s) returned
✓ 53	21:27:07	SELECT COUNT(*) AS Total_Transactions ...	1 row(s) returned
✓ 54	21:27:24	SELECT SUM(amount) AS Total_Deposit F...	1 row(s) returned
✓ 55	21:27:44	SELECT branch_name, SUM(amount) ...	3 row(s) returned

GROUP BY +HAVING

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

Administration Schemas Information

No object selected

```

38 GROUP BY branch_name;
39
40 • SELECT branch_name,
41       SUM(amount) AS Total_Amount
42 FROM bank_transactions
43 GROUP BY branch_name
44 HAVING SUM(amount) > 20000;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

branch_name	Total_Amount
Hyderabad	34500.00
Vijayawada	23000.00

Result 8 x

Output

Action Output

#	Time	Action	Message
51	21:26:22	SELECT MAX(amount) AS Highest_Transac...	1 row(s) returned
52	21:26:47	SELECT MIN(amount) AS Lowest_Transac...	1 row(s) returned
53	21:27:07	SELECT COUNT(*) AS Total_Transactions ...	1 row(s) returned
54	21:27:24	SELECT SUM(amount) AS Total_Deposit F...	1 row(s) returned
55	21:27:44	SELECT branch_name, SUM(amount) ...	3 row(s) returned
56	21:28:09	SELECT branch_name, SUM(amount) ...	2 row(s) returned

GROUP BY + ORDER BY

SCHEMAS
 Filter objects
 ▶ bankdb
 ▶ bookflow_db
 ▶ bookflow_week2_db
 ▶ college_yt
 ▶ practice
 ▶ sql_clg
 ▶ sys

Limit to 1000 rows
 44 **HAVING** SUM(amount) > 20000;
 45
 46 • **SELECT** branch_name,
 47 SUM(amount) **AS** Total_Amount
 48 **FROM** bank_transactions
 49 **GROUP BY** branch_name
 50 **ORDER BY** Total_Amount **DESC**;

Result Grid
 Filter Rows:
 Export: Wrap Cell Content:

	branch_name	Total_Amount
▶	Hyderabad	34500.00
	Vijayawada	23000.00
	Vizag	12000.00

Administration Schemas
 Information
 No object selected

Result 9 ×
 Output
 Action Output

#	Time	Action	Message	Dur
✓ 52	21:26:47	SELECT MIN(amount) AS Lowest_Transac...	1 row(s) returned	0.00
✓ 53	21:27:07	SELECT COUNT(*) AS Total_Transactions ...	1 row(s) returned	0.01
✓ 54	21:27:24	SELECT SUM(amount) AS Total_Deposit F...	1 row(s) returned	0.00
✓ 55	21:27:44	SELECT branch_name, SUM(amount) ...	3 row(s) returned	0.00
✓ 56	21:28:09	SELECT branch_name, SUM(amount) ...	2 row(s) returned	0.00
✓ 57	21:28:25	SELECT branch_name, SUM(amount) ...	3 row(s) returned	0.00

GROUP BY + HAVING+ORDER BY

HAVING + COUNT()

SCHEMAS

Filter objects

bankdb

bookflow_db

bookflow_week2_db

college_yt

practice

sql_dlg

sys

Administration

Schemas

Information

No object selected

Object Info

Session

Limit to 1000 rows

56 HAVING COUNT(*) >= 3

57 ORDER BY Total_Transactions DESC;

58 SELECT branch_name,

59 COUNT(*) AS Total_Transactions

60 FROM bank_transactions

61 GROUP BY branch_name

62 HAVING COUNT(*) > 3;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

branch_name	Total_Transactions
Hyderabad	5

Result 11 x

Output

Action Output

#	Time	Action	Message
✓ 54	21:27:24	SELECT SUM(amount) AS Total_Deposit F...	1 row(s) returned
✓ 55	21:27:44	SELECT branch_name, SUM(amount) ...	3 row(s) returned
✓ 56	21:28:09	SELECT branch_name, SUM(amount) ...	2 row(s) returned
✓ 57	21:28:25	SELECT branch_name, SUM(amount) ...	3 row(s) returned
✓ 58	21:28:58	SELECT branch_name, COUNT(*) AS ...	2 row(s) returned
✓ 59	21:29:45	SELECT branch_name, COUNT(*) AS ...	1 row(s) returned

GROUP BY + AVG()

Navigator: SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

Administration Schemas

Information

No object selected

Object Info Session

SQL File 6

```

60 FROM bank_transactions
61 GROUP BY branch_name
62 HAVING COUNT(*) > 3;
63 SELECT branch_name,
64        AVG(amount) AS Avg_Amount
65 FROM bank_transactions
66 GROUP BY branch_name;

```

Result Grid

branch_name	Avg_Amount
Hyderabad	6900.000000
Vijayawada	7666.666667
Vizag	6000.000000

Result 12

Output

Action Output

#	Time	Action	Message
55	21:27:44	SELECT branch_name, SUM(amount) ...	3 row(s) returned
56	21:28:09	SELECT branch_name, SUM(amount) ...	2 row(s) returned
57	21:28:25	SELECT branch_name, SUM(amount) ...	3 row(s) returned
58	21:28:58	SELECT branch_name, COUNT(*) AS ...	2 row(s) returned
59	21:29:45	SELECT branch_name, COUNT(*) AS ...	1 row(s) returned
60	21:30:09	SELECT branch_name, AVG(amount) ...	3 row(s) returned

WHERE + GROUP BY + HAVING

SCHEMAS

Filter objects

- bankdb
- bookflow_db
- bookflow_week2_db
- college_yt
- practice
- sql_clg
- sys

```
66 GROUP BY branch_name;
67 • SELECT branch_name,
68     SUM(amount) AS Total_Deposit
69 FROM bank_transactions
70 WHERE transaction_type = 'Deposit'
71 GROUP BY branch_name
72 HAVING SUM(amount) > 15000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	branch_name	Total_Deposit
▶	Hyderabad	29000.00
	Vijayawada	22000.00

Administration | Schemas | Information

No object selected

Result 13 x

Output

Action Output

#	Time	Action	Message
✓ 56	21:28:09	SELECT branch_name, SUM(amount) ...	2 row(s) returned
✓ 57	21:28:25	SELECT branch_name, SUM(amount) ...	3 row(s) returned
✓ 58	21:28:58	SELECT branch_name, COUNT(*) AS ...	2 row(s) returned
✓ 59	21:29:45	SELECT branch_name, COUNT(*) AS ...	1 row(s) returned
✓ 60	21:30:09	SELECT branch_name, AVG(amount) ...	3 row(s) returned
✓ 61	21:30:33	SELECT branch_name, SUM(amount) ...	2 row(s) returned

GROUP BY + HAVING + ORDER BY

WEEK1

001

WEEK2

SQL File 5

SQL File 6

SQL File 7

Limit to 1000 rows

73

74 • `SELECT branch_name,`

75 `AVG(amount) AS Avg_Amount`

76 `FROM bank_transactions`

77 `GROUP BY branch_name`

78 `HAVING AVG(amount) > 7000`

79 `ORDER BY Avg_Amount DESC;`

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

branch_name	Avg_Amount
Vijayawada	7666.666667

Administration

Schemas

Information

No object selected

Object Info

Session

Result 14

Output

Action Output

#	Time	Action	Message
✓ 57	21:28:25	SELECT branch_name, SUM(amount) ...	3 row(s) returned
✓ 58	21:28:58	SELECT branch_name, COUNT(*) AS ...	2 row(s) returned
✓ 59	21:29:45	SELECT branch_name, COUNT(*) AS ...	1 row(s) returned
✓ 60	21:30:09	SELECT branch_name, AVG(amount) ...	3 row(s) returned
✓ 61	21:30:33	SELECT branch_name, SUM(amount) ...	2 row(s) returned
✓ 62	21:31:09	SELECT branch_name, AVG(amount) ...	1 row(s) returned